

Mr. MARCONI. The wave length there is 7,000 meters.

Senator SMITH. When was that communication between Ireland and Buenos Aires?

Mr. MARCONI. It was in October, in 1910.

Senator SMITH. Is there any proficiency test prescribed by any special board in England?

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Mr. MARCONI. Yes; the operators have to pass a proficiency test before the post office authorities, which control the telegraphs in England.

Senator SMITH. Is there any in this country?

Mr. MARCONI. I believe there is now.

Senator SMITH. How recently?

Mr. MARCONI. Since a law was passed compelling passenger-carrying vessels to carry wireless-telegraph apparatus.

Senator SMITH. About two years ago?

Mr. MARCONI. About two years ago.

Senator SMITH. There seems to be a distinction between commercial business and distress or emergency business, ships business. Why should that be so?

Mr. MARCONI. For this reason: The commercial business is paid for and accounted for between the ships and the shore stations and organizations working the telegraphs on land, whilst, of course, for distress messages and messages affecting the safety of ships no charge is made and is not in itself a commercial business.

Senator SMITH. The Berlin convention, however, rather exalts the emergency phase of wireless telegraphy, giving to distress calls the precedence over all other calls, does it not?

Mr. MARCONI. Yes, sir; it has copied us in that, because that was one of our provisions before there was any Berlin convention.

Senator SMITH. It even takes precedence of Government business, does it not?

Mr. MARCONI. Even of Government business; yes.

Senator SMITH. Mr. Marconi, were any orders given by the Marconi Co., to the operators or the operator on the *Carpathia*, with reference to the receipt and answer of messages?

Mr. MARCONI. None whatever.

Senator SMITH. Do you know anything about the effort of the President of the United States to communicate with the *Carpathia*?

Mr. MARCONI. Except what I have read in the newspapers.

Senator SMITH. But, so far as you know, there was no disposition to censorize or control the operator of the *Carpathia*.

Mr. MARCONI. There was none whatever; and further, I was very much surprised at the things that were stated in the press, that a reply had been refused or had not been transmitted.

Senator SMITH. Do you know whether there was a reply refused?

Mr. MARCONI. Only from what I saw in the press. I might say that the operator, of course, can speak for himself; but I asked him that question last night when I boarded the *Carpathia* and he told me that he never dreamed of refusing to reply to a message sent by the President.

Senator SMITH. I think that is all. We are very much obliged to you.

TESTIMONY OF CHARLES HERBERT LIGHTOLLER.

Mr. Lightoller was sworn by the chairman.
Senator SMITH. What is your name?
Mr. LIGHTOLLER. Charles Herbert Lightoller.
Senator SMITH. Mr. Lightoller, where do you reside?
Mr. LIGHTOLLER. Netley Abbey, Hampshire.
Senator SMITH. England?

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Mr. LIGHTOLLER. England.
Senator SMITH. How old are you?
Mr. LIGHTOLLER. Thirty-eight.
Senator SMITH. What is your business?
Mr. LIGHTOLLER. Seaman.
Senator SMITH. How long have you been in the service or employment?
Mr. LIGHTOLLER. Thirteen years and three months.
Senator SMITH. How extensive has been your service in that time?
Mr. LIGHTOLLER. I do not quite follow you.
Senator SMITH. How much service have you seen? In what capacities?
Mr. LIGHTOLLER. In all the capacities in the White Star service - fourth, third, second, and first officer.
Senator SMITH. You have been in the White Star service during all of that time?
Mr. LIGHTOLLER. Yes, sir.
Senator SMITH. What official positions have you held?
Mr. LIGHTOLLER. Fourth, third, second, and first officer.
Senator SMITH. What position do you occupy now?
Mr. LIGHTOLLER. Second officer of the *Titanic*.
Senator SMITH. How long have you been second officer?
Mr. LIGHTOLLER. Altogether, about seven years.
Senator SMITH. When did you go aboard the *Titanic*?
Mr. LIGHTOLLER. In Belfast.
Senator SMITH. When?
Mr. LIGHTOLLER. March 19 or 20.
Senator SMITH. Did you make the so-called trial trips?
Mr. LIGHTOLLER. Yes, sir.
Senator SMITH. Of what did they consist?
Mr. LIGHTOLLER. Turning circles and adjusting compasses.
Senator SMITH. In what waters?
Mr. LIGHTOLLER. Belfast Lough.
Senator SMITH. How extensive is that lough?
Mr. LIGHTOLLER. I can hardly say offhand without seeing a chart.
Senator SMITH. Have you any data here that shows?
Mr. LIGHTOLLER. No.
Senator SMITH. Just state as nearly as you can.
Mr. LIGHTOLLER. It may be about 15 miles long, widening out from a few miles wide to perhaps 7 miles. That is only approximate, sir.

Senator SMITH. Have you ever been in that water before?

Mr. LIGHTOLLER. Only passing through.

Senator SMITH. How did you happen to pass through it?

Mr. LIGHTOLLER. Going into Belfast or coming out bound to some port. I do not mean in an official capacity; as a passenger. I have been through it in an official capacity about 11 years ago.

Senator SMITH. Is that water usually selected for these trial tests for new ships?

Mr. LIGHTOLLER. Yes.

Senator SMITH. What was the condition of the weather when you made this trial test?

Mr. LIGHTOLLER. Light breeze, clear weather, sir.

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Senator SMITH. From the time you boarded the *Titanic* did you at any time encounter any rough weather?

Mr. LIGHTOLLER. No, sir.

Senator SMITH. You were always in smooth water, so called?

Mr. LIGHTOLLER. Yes, sir.

Senator SMITH. Does that include up to the time of this collision?

Mr. LIGHTOLLER. Yes, sir.

Senator SMITH. Of what do these trial tests consist?

Mr. LIGHTOLLER. Turning circles.

Senator SMITH. I wish you would describe that a little more fully. Under what head of steam and how fast would the boat be moving?

Mr. LIGHTOLLER. Under various speeds.

Senator SMITH. In how large a radius would these circles be made?

Mr. LIGHTOLLER. Turning circles consists of seeing in what space the ship will turn under certain helms with the engines at various speeds.

Senator SMITH. Was this boat tested at its maximum speed?

Mr. LIGHTOLLER. That I could not say, sir.

Senator SMITH. What was the maximum speed of this boat?

Mr. LIGHTOLLER. That I could not say, sir. She was never put, to my knowledge, to her maximum speed.

Senator SMITH. What did you understand it to be?

Mr. LIGHTOLLER. About 22 1/2 to 23 knots.

Senator SMITH. From whom did you get that information?

Mr. LIGHTOLLER. General rumor, sir.

Senator SMITH. Did you talk with the boat's officers?

Mr. LIGHTOLLER. From talk generally; yes. It was only an approximate idea.

Senator SMITH. How much time was spent in the test?

Mr. LIGHTOLLER. I could not say exactly.

Senator SMITH. Approximately?

Mr. LIGHTOLLER. About five hours.

Senator SMITH. During that time those circles were made?

Mr. LIGHTOLLER. Yes, sir.

Senator SMITH. And the ship reversed?

Mr. LIGHTOLLER. Yes, sir.

Senator SMITH. And put on a straight course?

Mr. LIGHTOLLER. Yes, sir.

Senator SMITH. And under full head?

Mr. LIGHTOLLER. I could not say, sir. She steamed for a certain distance under approximately a full head of steam; but how much steam was on I could not say, or what pressure of steam.

Senator SMITH. How many engines were there in this boat?

Mr. LIGHTOLLER. Two reciprocating and one turbine.

Senator SMITH. Were they all working on the trial test?

Mr. LIGHTOLLER. So far as I know, sir.

Senator SMITH. What do you know about that? Were you in the engine room?

Mr. LIGHTOLLER. No, sir; I was on my station, aft.

Senator SMITH. Where was your station?

Mr. LIGHTOLLER. The after end of the ship.

Senator SMITH. Then you would not, of your own knowledge, know whether its entire power was being tested out or not?

Mr. LIGHTOLLER. I should not; no, sir.

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Senator SMITH. Five hours was the length of time spent in making those tests?

Mr. LIGHTOLLER. Approximately the length of time occupied in turning those circles.

Senator SMITH. What was the next thing that was done with the ship?

Mr. LIGHTOLLER. She was run a certain distance on a comparatively straight course and back again.

Senator SMITH. How far?

Mr. LIGHTOLLER. I could not say without a chart, sir.

Senator SMITH. How long did it take you?

Mr. LIGHTOLLER. Approximately four hours.

Senator SMITH. To make the straight run?

Mr. LIGHTOLLER. Yes, sir.

Senator SMITH. And return?

Mr. LIGHTOLLER. And return.

Senator SMITH. Four hours all together, two out and two back?

Mr. LIGHTOLLER. Two out and two back. That is only approximate.

Senator SMITH. Would you think from what you observed in the movements of this ship that it was going pretty fast?

Mr. LIGHTOLLER. For a ship of that size, a fair speed.

Senator SMITH. Fair speed?

Mr. LIGHTOLLER. A fair speed.

Senator SMITH. What would you call real good speed?

Mr. LIGHTOLLER. When the ship was built, we only expected her to go 21 knots, therefore all over 21 we thought very good.

Senator SMITH. This ship exceeded 21 knots?

Mr. LIGHTOLLER. On the trials? I am not speaking of the trials. I do not know what the speed was; I have no idea.

Senator SMITH. I understand you to say that you expected to get 21 knots out of her?

Mr. LIGHTOLLER. The builders, I presume, to get 21.

Senator SMITH. That was the general rumor?

Mr. LIGHTOLLER. Yes.

Senator SMITH. Among the officers?

Mr. LIGHTOLLER. Yes.

Senator SMITH. I suppose that was the hope, too, of the officers?

Mr. LIGHTOLLER. Exactly.

Senator SMITH. What boat had you been on before you went on board the *Titanic*?

Mr. LIGHTOLLER. The *Oceanic*.

Senator SMITH. The *Oceanic*, of the same line?

Mr. LIGHTOLLER. Of the same line.

Senator SMITH. How large a boat is the *Oceanic*?

Mr. LIGHTOLLER. Seventeen thousand tons gross.

Senator SMITH. Do you know her maximum speed?

Mr. LIGHTOLLER. Twenty-one knots.

Senator SMITH. I want to be sure I get the results of these trial tests accurately. I want you to tell me how long it took to make these tests. The straightaway tests and the circle tests altogether consumed how much time?

Mr. LIGHTOLLER. Approximately six or seven hours. I could not say any nearer than that.

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Senator SMITH. What time of day did you begin these tests?

Mr. LIGHTOLLER. In the morning.

Senator SMITH. How early?

Mr. LIGHTOLLER. About 10 o'clock.

Senator SMITH. Was it clear weather?

Mr. LIGHTOLLER. Perfectly clear.

Senator SMITH. Was there any sea?

Mr. LIGHTOLLER. Very little.

Senator SMITH. And after about seven hours the tests were concluded?

Mr. LIGHTOLLER. With the exception of full speed astern; that is to see in what distance the ship will stop with the engines full speed astern - what we call the full speed astern test.

Senator SMITH. Was that made that day?

Mr. LIGHTOLLER. Yes, sir.

Senator SMITH. How long did that take?

Mr. LIGHTOLLER. That was only the matter of minutes.

Senator SMITH. A few minutes?

Mr. LIGHTOLLER. A few minutes.

Senator SMITH. Do you know who was aboard the *Titanic* in these trial tests?

Mr. LIGHTOLLER. A great number. I know some of them.

Senator SMITH. Please state those that you know.

Mr. LIGHTOLLER. Capt. Smith; Mr. Murdoch, chief officer; myself, first officer; Mr. Blair, second officer; Mr. Pitman, third officer; Mr. Boxhall, fourth officer; Mr. Lowe, fifth officer; Mr. Moody, sixth officer; and Mr. Andrews, of Harland & Wolff.

Senator SMITH. Representing the builders?

Mr. LIGHTOLLER. Yes, sir. I could not say anyone else with any accuracy.

Senator SMITH. Who was the chief engineer?

Mr. LIGHTOLLER. Mr. Bell, chief engineer; Mr. Ferguson, second engineer; Mr. Hesketh, also second. That is all I know.

Senator SMITH. How many men constituted the crew?

Mr. LIGHTOLLER. Seamen, you are speaking of?

Senator SMITH. Yes.

Mr. LIGHTOLLER. About 71 all told; officers and crew.

Senator SMITH. And seamen?

Mr. LIGHTOLLER. Yes, sir.

Senator SMITH. On the trial test?

Mr. LIGHTOLLER. Oh, no, sir. I am not speaking of the trial.

Senator SMITH. How many men constituted the crew on the trial tests?

Mr. LIGHTOLLER. About 30 of the crew and about 30 of what we call runners.

Senator SMITH. Were there any guests on the boat?

Mr. LIGHTOLLER. I believe there were; I could not say who.

Senator SMITH. Do you know who they were?

Mr. LIGHTOLLER. No, sir.

Senator SMITH. Were there any of the officers of the White Star Line?

Mr. LIGHTOLLER. I could not say with certainty, sir.

Senator SMITH. You do not recall seeing any of them?

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Mr. LIGHTOLLER. I do not recall; no, sir. I believe there were some on board; but I can not remember who they were. I was not brought in contact with them.

Senator SMITH. Was Mr. Ismay aboard?

Mr. LIGHTOLLER. Not to my knowledge, sir.

Senator SMITH. Did you hear afterwards that he was on board?

Mr. LIGHTOLLER. No, sir.

Senator SMITH. You can not recall any officer of the company that was?

Mr. LIGHTOLLER. No, sir.

Senator SMITH. I mean any general officer?

Mr. LIGHTOLLER. No, sir.

Senator SMITH. Or director?

Mr. LIGHTOLLER. No, sir.

Senator SMITH. Was there anybody aboard representing the British Government?

Mr. LIGHTOLLER. Not to my knowledge, sir.

Senator SMITH. Were there any other officers of any other White Star Line boats?

Mr. LIGHTOLLER. No, sir.

Senator SMITH. After the final test, what was done with the boat?

Mr. LIGHTOLLER. We proceeded toward Southampton.

Senator SMITH. Immediately?

Mr. LIGHTOLLER. Almost immediately after taking on board a few things that had been left behind, which were required for the completion of the ship.

Senator SMITH. What?

Mr. LIGHTOLLER. So far as I know, requisites down in the galley, cooking apparatus, a few chairs, and such things as that.

Senator SMITH. Was the life-saving equipment -

Mr. LIGHTOLLER. Oh, no, sir; nothing like that.

Senator SMITH. Was the life-saving equipment complete?

Mr. LIGHTOLLER. Yes, sir.

Senator SMITH. Of what did it consist?

Mr. LIGHTOLLER. The necessary number of lifeboats.

Senator SMITH. I wish you would say how that is determined, if you can.

Mr. LIGHTOLLER. By the number of people on board.

Senator SMITH. You do not know how many there are on board until you are ready to start?

Mr. LIGHTOLLER. No sir.

Senator SMITH. Is it not determined by the number of accommodations rather than by the number of people who get aboard?

Mr. LIGHTOLLER. There must be life-saving apparatus for every one on board, regardless of accommodations.

Senator SMITH. Yes; but what I desire to know is whether in each stateroom on each deck, in all classes, whether there is any rule, and whether it was followed at that time, so far as you know, in equipping this boat with life preservers and life belts and anything else that might appropriately go into the rooms and be upon the decks of a boat of that character?

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Mr. LIGHTOLLER. She was perfectly complete throughout, sir.

Senator SMITH. How many lifeboats were there?

Mr. LIGHTOLLER. Sixteen.

Senator SMITH. All of the same type?

Mr. LIGHTOLLER. Consisting of 14 lifeboats, 2 emergency boats, and 4 collapsible boats.

Senator SMITH. Tell us whether they were new entirely.

Mr. LIGHTOLLER. Entirely new.

Senator SMITH. And in their proper places?

Mr. LIGHTOLLER. In their proper places.

Senator SMITH. With the necessary lowering apparatus?

Mr. LIGHTOLLER. Everything complete, examined by the officers of the ship.

Senator SMITH. Was a test of the lifeboats made before you sailed for Southampton?

Mr. LIGHTOLLER. All the gear was tested.

Senator SMITH. Were the lifeboats lowered?

Mr. LIGHTOLLER. Yes, sir.

Senator SMITH. Under whose orders?

Mr. LIGHTOLLER. The officers, principally my orders.

Senator SMITH. Under your orders?

Mr. LIGHTOLLER. Yes.

Senator SMITH. Did you see the work done?

Mr. LIGHTOLLER. I did.

Senator SMITH. Tell just what was done.

Mr. LIGHTOLLER. All the boats on the ship were swung out and those that I required were lowered

down as far as I wanted them - some all the way down, and some dropped into the water.

Senator SMITH. I wish you would give the proportion that went into the water.

Mr. LIGHTOLLER. About six.

Senator SMITH. Six into the water?

Mr. LIGHTOLLER. Yes.

Senator SMITH. And the others lowered?

Mr. LIGHTOLLER. Part of the way - as far as I thought necessary.

Senator SMITH. Part of the way?

Mr. LIGHTOLLER. Yes.

Senator SMITH. Of course, part of the way would not do anybody much good on a sinking ship. I assume you did that for the purpose of trying the gear, and not for the purpose of testing the security of the lifeboats?

Mr. LIGHTOLLER. It is principally the gear that we test. The lifeboats we know to be all right.

Senator SMITH. These boats were lowered from what deck?

Mr. LIGHTOLLER. From the boat deck.

Senator SMITH. Is that the sun deck?

Mr. LIGHTOLLER. That is the top deck.

Senator SMITH. Do you know how far it was from that top deck to the water?

Mr. LIGHTOLLER. Seventy feet.

Senator SMITH. What time did you reach Southampton?

Mr. LIGHTOLLER. About midnight.

Senator SMITH. Of what night?

Mr. LIGHTOLLER. I could not say.

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Senator SMITH. Think it over.

Mr. LIGHTOLLER. I think it was the morning of the 4th of April.

Senator SMITH. What makes you think it was the morning of the 4th?

Mr. LIGHTOLLER. Because we sailed on the 10th.

Senator SMITH. How long did it take to make the run to Southampton?

Mr. LIGHTOLLER. About 24 hours.

Senator SMITH. Did you strike any heavy weather?

Mr. LIGHTOLLER. No, sir.

Senator SMITH. How fast did you go?

Mr. LIGHTOLLER. About 18 knots.

Senator SMITH. What was done when you reached Southampton?

Mr. LIGHTOLLER. The ship was heeled for stability.

Senator SMITH. Just describe that.

Mr. LIGHTOLLER. The builders knowing the exact weights on board, additional weights are placed on each side of the ship. A pendulum is suspended in the most convenient place in the ship with a plumb on the end of it, and a method of registering the difference with the plumb line; a number of men then transfer the weights from one side of the ship to the other, bringing all the weight on one side and transferring the whole of it back again; and with this, I believe the builders are able to draw up a stability scale.

Senator SMITH. From what part of the ship are these tests made?

Mr. LIGHTOLLER. The weights carried over, you mean?
Senator SMITH. Yes. From the upper part?
Mr. LIGHTOLLER. The "C" deck - the third deck down.
Senator SMITH. About the center of the ship?
Mr. LIGHTOLLER. Not quite the center of the ship.
Senator SMITH. Were there any tests made from the upper deck?
Mr. LIGHTOLLER. Not that I know of, sir.
Senator SMITH. What else was done at Southampton?
Mr. LIGHTOLLER. We shipped coal, provisions, cargo was taken on board, passed the Board of Trade tests and survey.
Senator SMITH. Did some British officer make the Board of Trade test?
Mr. LIGHTOLLER. The Southampton Board of Trade officer.
Senator SMITH. What did he do?
Mr. LIGHTOLLER. He carried out the requisite tests required by the Board of Trade.
Senator SMITH. Did you accompany him?
Mr. LIGHTOLLER. Yes; I was with him part of the time.
Senator SMITH. Who was this officer of the British Board of Trade?
Mr. LIGHTOLLER. Capt. Clark.
Senator SMITH. He was an officer?
Mr. LIGHTOLLER. He was purely a representative of the British Board of Trade, appointed by the British Board of Trade, with post at the port of Southampton; surveyor.
Senator SMITH. He was assigned to Southampton?
Mr. LIGHTOLLER. Yes, sir.
Senator SMITH. How old a man was he?
Mr. LIGHTOLLER. About 45.
Senator SMITH. Of English nationality?
Mr. LIGHTOLLER. Yes.

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Senator SMITH. Had you ever seen him before?
Mr. LIGHTOLLER. Frequently.
Senator SMITH. Do you know whether he had any experience in marine service?
Mr. LIGHTOLLER. All surveyors, I understand, have been in command. I know he had for a number of years.
Senator SMITH. What does that mean - that he had been "in command"?
Mr. LIGHTOLLER. In command of a British ship; captain.
Senator SMITH. How much time did this officer spend on the ship?
Mr. LIGHTOLLER. That I could not say, sir.
Senator SMITH. How much did he spend when he was with you?
Mr. LIGHTOLLER. About four hours.
Senator SMITH. Then did you turn him over to some other officer?
Mr. LIGHTOLLER. Yes, sir.
Senator SMITH. To what other officer?
Mr. LIGHTOLLER. I think it was the first.
Senator SMITH. What is his name?

Mr. LIGHTOLLER. Mr. Murdoch.

Senator SMITH. Did he survive the *Titanic* disaster?

Mr. LIGHTOLLER. No, sir. He was chief then.

Senator SMITH. He did not survive?

Mr. LIGHTOLLER. No, sir.

Senator SMITH. Do you know whether any other officer of the ship accompanied this inspector during his stay on board?

Mr. LIGHTOLLER. That I could not say with certainty.

Senator SMITH. What is your best judgment about it?

Mr. LIGHTOLLER. I should say the marine superintendent was with him the whole time.

Senator SMITH. The marine superintendent?

Mr. LIGHTOLLER. Of the White Star Line, at Southampton.

Senator SMITH. What is his name?

Mr. LIGHTOLLER. Capt. Steele.

Senator SMITH. How old a man is he?

Mr. LIGHTOLLER. About 50.

Senator SMITH. Is he a commander?

Mr. LIGHTOLLER. Yes, sir.

Senator SMITH. Had you ever seen one of those ocean liners inspected by the British Board of Trade representative before?

Mr. LIGHTOLLER. Frequently.

Senator SMITH. How thorough are they about it?

Mr. LIGHTOLLER. Speaking of Capt. Clark, we call him a nuisance because he is so strict.

Senator SMITH. Capt. Clark?

Mr. LIGHTOLLER. Yes, sir.

Senator SMITH. Is he the marine officer?

Mr. LIGHTOLLER. That is the Board of Trade representative.

Senator SMITH. In what respect is he a nuisance?

Mr. LIGHTOLLER. Because he makes us fork out every detail.

Senator SMITH. I should suppose you would be quite willing to do that?

Mr. LIGHTOLLER. Perfectly willing.

Senator SMITH. Do you mean by that that he would call attention to the absence of tools, implements, and devices necessary for the ship's full equipment?

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Mr. LIGHTOLLER. No, sir. He would insist upon them all being absolutely brought out on deck every time.

Senator SMITH. On what?

Mr. LIGHTOLLER. Everything that contributes to the ship's equipment.

Senator SMITH. What would that consist of?

Mr. LIGHTOLLER. The whole of the ship's life-saving equipment.

Senator SMITH. Life preservers?

Mr. LIGHTOLLER. Life preservers throughout the ship, all the boats turned out, uncovered, all the tanks examined, all the breakers examined, oars counted, boats turned out, rudders tried, all the davits tried - there was innumerable detail work.

Senator SMITH. And the boats lowered?

Mr. LIGHTOLLER. The boats lowered, put in the water, and pulled out, and brought back again, and if he was not satisfied, sent back again.

Senator SMITH. And the ropes and chains tested?

Mr. LIGHTOLLER. Yes, sir.

Senator SMITH. When he inspected your ship, about where would he find these life preservers?

Mr. LIGHTOLLER. Life belts in every room, in every compartment, where, as we say, there was habitation, where a man could live.

Senator SMITH. Would that include the steerage?

Mr. LIGHTOLLER. Yes, undoubtedly; and the crews quarters.

Senator SMITH. In the steerage do they have rooms?

Mr. LIGHTOLLER. Yes, sir.

Senator SMITH. Are they equipped?

Mr. LIGHTOLLER. Yes, sir.

Senator SMITH. With the same apparatus for the preservation of life with an emergency as the first and second cabins?

Mr. LIGHTOLLER. Identically the same.

Senator SMITH. You used the term "life belt."

Mr. LIGHTOLLER. Yes, sir.

Senator SMITH. I wish you would describe a life belt.

Mr. LIGHTOLLER. It consists of a series of pieces of cork - allow me to show you by illustration - a hole is cut in there [illustrating] for the head to go through and this falls over front and back, and there are tapes from the back then tied around the front. It is a new idea and very effective, because no one can make a mistake in putting it on.

Senator SMITH. Is there cork on both sides?

Mr. LIGHTOLLER. On both sides.

Senator SMITH. Are the arms free?

Mr. LIGHTOLLER. Free, absolutely.

Senator SMITH. And when in the water does this adhere or extend?

Mr. LIGHTOLLER. It is tied to the body.

Senator SMITH. It is tied to the body?

Mr. LIGHTOLLER. Yes, sir.

Senator SMITH. Have you ever had one of these on?

Mr. LIGHTOLLER. Yes, sir.

Senator SMITH. Have you ever been into the sea with one of them?

Mr. LIGHTOLLER. Yes, sir.

Senator SMITH. Where?

Mr. LIGHTOLLER. From the *Titanic*.

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Senator SMITH. In this recent collision?

Mr. LIGHTOLLER. Yes, sir.

Senator SMITH. How long were you in the sea with a life belt on?

Mr. LIGHTOLLER. Between half an hour and an hour.

Senator SMITH. What time did you leave the ship?